

**SALINA, KS, USA**WMO: **724586**Lat: **38.800N**Lon: **97.650W**Elev: **387**StdP: **96.76**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **03919****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.2      | -12.6 | -20.6                         | 0.6 | -13.6 | -18.1 | 0.8 | -11.2 | 14.3                  | 1.5  | 12.6 | 2.6  | 4.3                   | 330  | 0.602 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.5                   | 38.7            | 23.3 | 37.0 | 23.3 | 35.2 | 23.0 | 25.3                | 33.6 | 24.7 | 32.9 | 24.0 | 32.1 | 7.0                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.0                            | 18.6 | 28.8 | 22.4 | 17.9 | 28.4 | 21.8 | 17.2 | 28.0 | 79.5          | 33.6 | 76.6 | 32.8 | 74.1 | 32.0 | 30.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 12.7          | 11.4 | 10.3 | -19.3                      | 41.5 | 2.9                | 1.5 | -21.4                                              | 42.5 | -23.0      | 43.4 | -24.7      | 44.2 | -26.7      | 45.3 |    |    |
| (5)               |      |      | -19.7                      | 26.9 | 2.7                | 1.0 | -21.7                                              | 27.6 | -23.3      | 28.2 | -24.8      | 28.8 | -26.8      | 29.6 |    |    |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 13.8   | -0.3  | 1.9   | 7.6   | 13.0  | 19.0  | 25.0  | 27.6  | 26.4  | 21.8  | 14.5  | 7.2   | 0.8   |  |  |
|                                                                        | DBStd       | 11.11  | 6.09  | 6.64  | 6.31  | 5.23  | 4.74  | 3.85  | 3.47  | 3.36  | 4.70  | 5.23  | 5.45  | 5.78  |  |  |
|                                                                        | HDD10.0     | 1127   | 321   | 236   | 118   | 28    | 1     | 0     | 0     | 0     | 1     | 18    | 116   | 288   |  |  |
|                                                                        | HDD18.3     | 2637   | 578   | 460   | 333   | 172   | 52    | 2     | 0     | 0     | 19    | 141   | 335   | 544   |  |  |
|                                                                        | CDD10.0     | 2504   | 1     | 9     | 45    | 118   | 280   | 451   | 547   | 509   | 353   | 157   | 32    | 3     |  |  |
|                                                                        | CDD18.3     | 973    | 0     | 0     | 2     | 12    | 72    | 203   | 289   | 251   | 122   | 21    | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 11350  | 0     | 2     | 37    | 161   | 740   | 2354  | 3649  | 2878  | 1296  | 222   | 12    | 0     |  |  |
|                                                                        | CDH26.7     | 5717   | 0     | 0     | 6     | 42    | 282   | 1187  | 2051  | 1506  | 577   | 65    | 2     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.8    | 4.6   | 4.9   | 5.3   | 5.7   | 4.8   | 5.0   | 4.5   | 4.1   | 4.5   | 4.7   | 4.7   | 4.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 806    | 17    | 25    | 60    | 75    | 107   | 114   | 96    | 85    | 71    | 62    | 35    | 24    |  |  |
|                                                                        | PrecMax     | 1475   | 73    | 70    | 275   | 178   | 211   | 255   | 455   | 349   | 306   | 172   | 118   | 79    |  |  |
|                                                                        | PrecMin     | 330    | 0     | 0     | 0     | 9     | 6     | 23    | 0     | 3     | 3     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 241    | 15    | 22    | 56    | 46    | 49    | 57    | 95    | 66    | 63    | 46    | 34    | 20    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 18.6  | 22.8  | 27.0  | 30.9  | 35.1  | 39.1  | 41.3  | 40.1  | 37.2  | 31.9  | 24.9  | 18.4  |  |  |
|                                                                        |             | MCWB   | 9.6   | 11.8  | 14.6  | 18.0  | 21.2  | 23.4  | 23.5  | 23.0  | 22.4  | 19.7  | 14.8  | 12.8  |  |  |
|                                                                        | 2%          | DB     | 14.8  | 18.7  | 23.8  | 27.5  | 32.1  | 36.9  | 39.3  | 38.0  | 34.4  | 28.2  | 21.6  | 15.0  |  |  |
|                                                                        |             | MCWB   | 7.5   | 10.6  | 14.6  | 17.3  | 21.0  | 23.3  | 23.5  | 23.1  | 21.9  | 18.1  | 13.3  | 9.1   |  |  |
|                                                                        | 5%          | DB     | 11.7  | 15.3  | 21.0  | 24.9  | 29.8  | 35.0  | 37.4  | 36.0  | 32.4  | 25.8  | 18.7  | 12.4  |  |  |
|                                                                        |             | MCWB   | 6.2   | 9.1   | 13.2  | 16.2  | 20.3  | 22.9  | 23.6  | 23.3  | 21.5  | 17.0  | 11.9  | 7.5   |  |  |
|                                                                        | 10%         | DB     | 8.7   | 12.3  | 18.1  | 22.5  | 27.6  | 32.9  | 35.4  | 33.8  | 30.1  | 23.1  | 16.2  | 9.6   |  |  |
|                                                                        |             | MCWB   | 4.6   | 7.2   | 11.8  | 15.1  | 19.7  | 22.3  | 23.5  | 22.9  | 20.6  | 16.2  | 11.1  | 5.5   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 10.8  | 14.3  | 18.6  | 20.5  | 23.8  | 25.7  | 26.3  | 26.1  | 24.5  | 21.9  | 16.5  | 14.2  |  |  |
|                                                                        |             | MCDB   | 16.3  | 18.4  | 24.5  | 27.0  | 30.8  | 34.5  | 35.3  | 34.3  | 31.6  | 27.4  | 21.5  | 16.9  |  |  |
|                                                                        | 2%          | WB     | 8.3   | 11.8  | 16.3  | 18.9  | 22.6  | 24.7  | 25.3  | 25.1  | 23.5  | 20.3  | 15.0  | 10.3  |  |  |
|                                                                        |             | MCDB   | 13.5  | 16.6  | 20.4  | 24.9  | 29.6  | 33.2  | 34.0  | 33.0  | 30.8  | 25.7  | 18.9  | 13.5  |  |  |
|                                                                        | 5%          | WB     | 6.5   | 9.6   | 14.6  | 17.6  | 21.5  | 23.9  | 24.8  | 24.4  | 22.7  | 18.7  | 13.3  | 7.7   |  |  |
|                                                                        |             | MCDB   | 11.5  | 15.4  | 19.6  | 23.4  | 27.9  | 32.2  | 33.3  | 32.2  | 29.9  | 23.7  | 18.1  | 11.9  |  |  |
|                                                                        | 10%         | WB     | 4.6   | 7.1   | 12.2  | 15.9  | 20.4  | 23.2  | 24.1  | 23.7  | 21.8  | 17.0  | 11.3  | 5.7   |  |  |
|                                                                        |             | MCDB   | 8.7   | 12.1  | 17.9  | 21.6  | 26.1  | 31.0  | 32.4  | 31.3  | 28.5  | 22.0  | 15.8  | 9.5   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.5  | 11.9  | 13.0  | 13.0  | 12.2  | 12.5  | 12.5  | 12.1  | 12.4  | 12.7  | 12.3  | 10.9  |  |  |
|                                                                        |             | MCDBR  | 16.8  | 17.3  | 17.6  | 16.6  | 14.8  | 14.7  | 14.9  | 14.8  | 14.3  | 15.9  | 16.2  | 15.5  |  |  |
|                                                                        |             | MCWBR  | 10.7  | 10.7  | 10.0  | 9.1   | 6.6   | 5.0   | 4.1   | 4.5   | 5.2   | 8.0   | 9.3   | 10.3  |  |  |
|                                                                        |             | MCDBR  | 16.1  | 16.0  | 14.0  | 14.3  | 12.9  | 13.3  | 12.8  | 12.6  | 12.3  | 12.3  | 13.5  | 14.3  |  |  |
|                                                                        | 5% WB       | MCWBR  | 10.6  | 10.5  | 9.1   | 8.9   | 6.6   | 5.5   | 4.6   | 5.0   | 5.4   | 7.3   | 8.7   | 10.1  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.284 | 0.294 | 0.328 | 0.362 | 0.393 | 0.409 | 0.419 | 0.415 | 0.377 | 0.340 | 0.295 | 0.279 |  |  |
|                                                                        | taud        |        | 2.503 | 2.475 | 2.412 | 2.344 | 2.332 | 2.323 | 2.325 | 2.331 | 2.385 | 2.463 | 2.518 | 2.508 |  |  |
|                                                                        | Ebn at Noon |        | 904   | 938   | 932   | 915   | 887   | 870   | 857   | 853   | 868   | 870   | 882   | 882   |  |  |
|                                                                        | Edh at Noon |        | 77    | 90    | 107   | 122   | 126   | 127   | 126   | 122   | 109   | 91    | 75    | 71    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.40  | 3.17  | 4.20  | 5.30  | 6.07  | 6.86  | 6.73  | 6.00  | 4.93  | 3.61  | 2.66  | 2.08  |  |  |
|                                                                        | RadStd      |        | 0.21  | 0.35  | 0.31  | 0.31  | 0.46  | 0.36  | 0.41  | 0.32  | 0.29  | 0.38  | 0.22  | 0.20  |  |  |

**Historical Trends**

|                              |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (21 neighbors) |       | (m)     |        |         |       |       |             |         |         |         |
|                              |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page